

Customer Segmentation & *A/B Testing* Suite

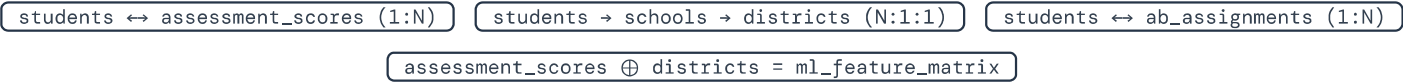
50

YEARS OF DATA HERITAGE

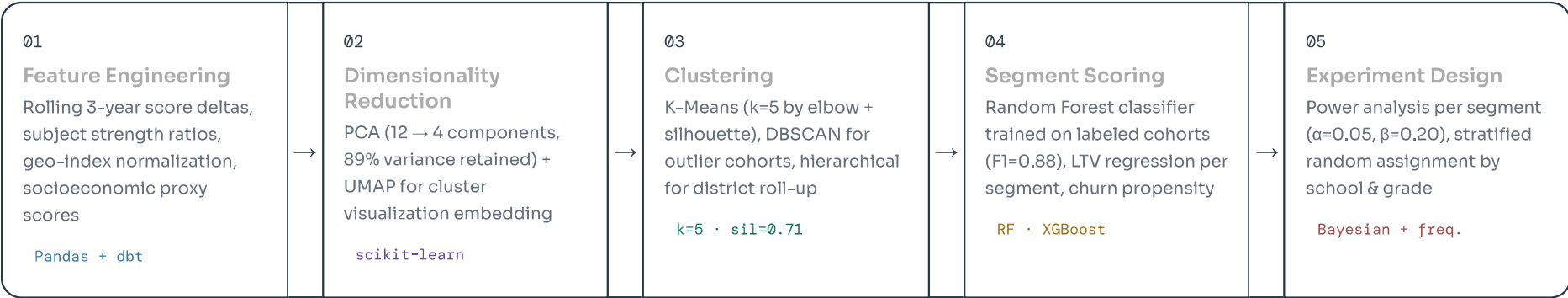
Personalized learning interventions driven by ML clustering, behavioral scoring, and controlled experimentation across 12,400+ schools in 47 districts. Grades 8–12.

ENTITY RELATIONSHIP MODEL — LINKED DATA ARCHITECTURE

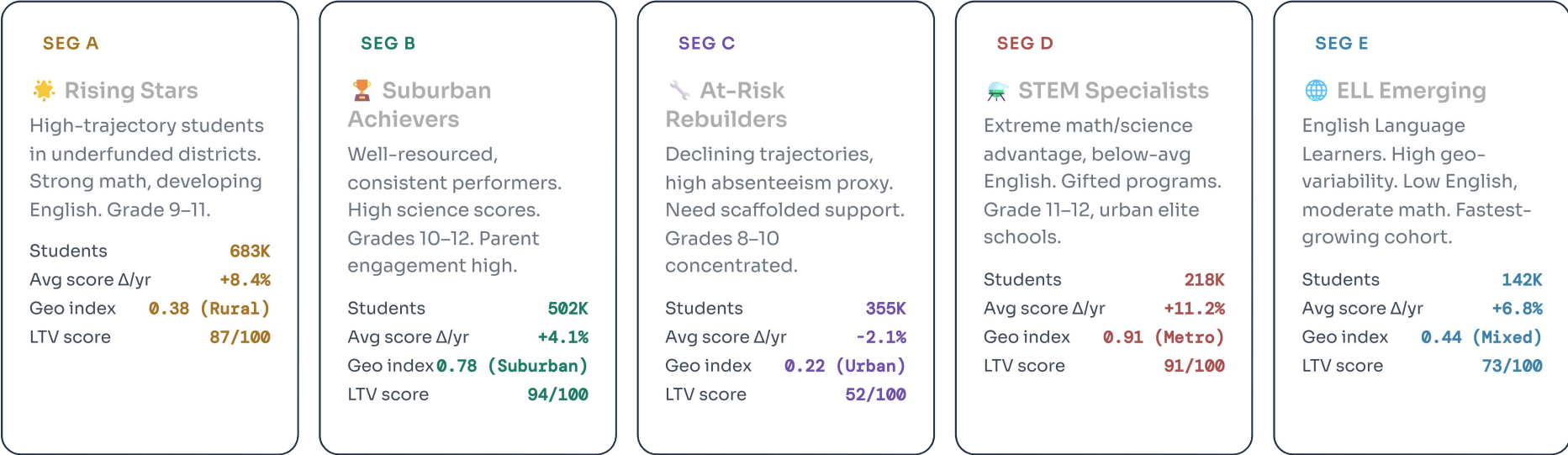
 students Core entity · fact table	 assessment_scores Measurement · time-series	 schools Dimension · geo context	 districts Dimension · policy level	 ab_assignments Experiment · derived table
student_id uuid PK school_id uuid FK cohort_id uuid FK grade int(8..12) age int gender_code char(1) first_enrolled date ell_status bool iep_flag bool free_lunch bool	score_id uuid PK student_id uuid FK test_date date subject enum raw_score float percentile float grade_equiv float geo_score float stanine int(1..9) is_makeup bool	school_id uuid PK district_id uuid FK school_name text school_type enum latitude float longitude float locale_code enum enrollment int title1_status bool avg_class_size float	district_id uuid PK district_name text state_code char(2) region enum funding_tier int(1..5) urbanicity enum tech_adoption float contract_start date arn text seats_licensed int	assignment_id uuid PK student_id uuid FK experiment_id uuid FK variant enum assigned_at timestamp segment_label char(1) pre_score float post_score float delta_score float converted bool Links via student_id → students and experiment_id → experiments



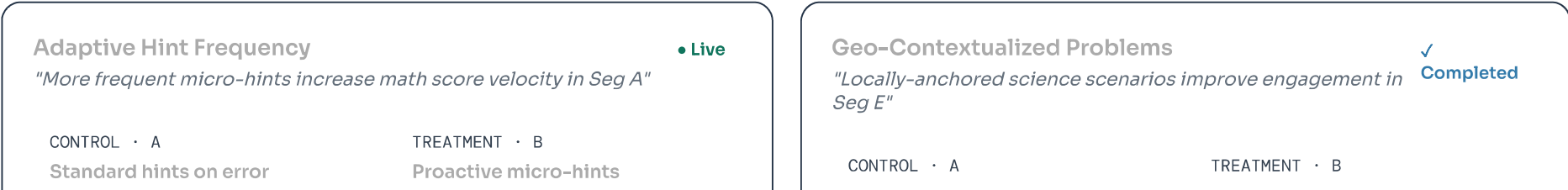
ML PIPELINE — FROM RAW SCORES TO SEGMENTS



5 IDENTIFIED CUSTOMER SEGMENTS — K-MEANS · N = 2.1M STUDENTS



ACTIVE & COMPLETED A/B EXPERIMENTS



Score Δ	+3.2%	Score Δ	+7.8%
Completion	71%	Completion	83%
n	12,440	n	12,381

+143% relative lift · $p < 0.001$ · 95% CI [3.8, 6.4]

Generic problems		Geo-contextualized	
Engagement	58%	Engagement	74%
Retention D30	44%	Retention D30	61%
n	8,220	n	8,195

+38% D30 retention lift · $p = 0.004$ · Rolled out to Seg E

Difficulty Auto-Scaling (DAS)

• Live

"IRT-based adaptive difficulty reduces frustration in Seg C"

CONTROL · A		TREATMENT · B	
Fixed curriculum path		IRT adaptive path	
Dropout rate	31%	Dropout rate	18%
Score Δ	-1.8%	Score Δ	+2.4%
n	9,110	n	9,084

-42% dropout reduction · $p = 0.018$ · Bayesian: 97.2% prob. superior

Teacher Dashboard Nudges

⌚ Planned Q3

"Weekly AI-generated insight nudges lift district renewal rates"

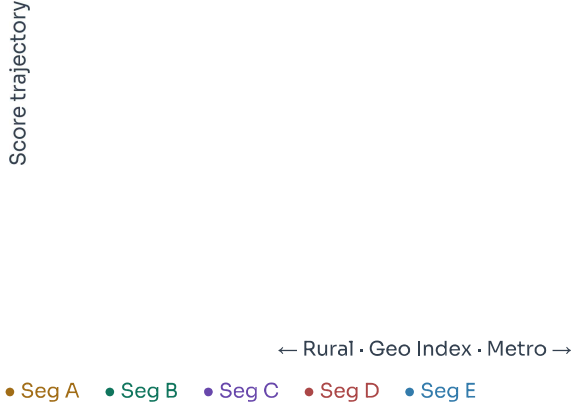
CONTROL · A		TREATMENT · B	
Standard reports		AI insight nudges	
Renewal est.	78%	Renewal est.	85%+
Login freq.	1.2×/wk	Login freq.	2.8×/wk
n (schools)	600	n (schools)	600

+7pp renewal target · Power: 85% · Launching Sep 2025

Avg subject scores by segment

English	72 A	88 B	58 C	66 D	49 E
Math	85 A	84 B	61 C	96 D	74 E
Science	78 A	92 B	55 C	98 D	68 E
Geo	64	81	48	73	71

Score trajectory vs geo-index (UMAP projection)



A

B

C

D

E

DISTRICT × SUBJECT PERFORMANCE HEATMAP — TOP 10 DISTRICTS

	Eng	Math	Sci	Geo	8th	9th	10th	11th	12th	Avg
Riverside USD	87	65	87	58	67	60	61	60	72	69
Oakland Unified	58	98	90	60	92	72	64	75	77	76
Mesa ISD	79	61	78	62	72	87	71	58	93	73
Fulton County	96	92	76	57	61	78	68	88	91	79
Denver PS	75	92	96	70	63	98	77	83	72	81
Atlanta City	44	52	40	44	53	81	74	84	80	61
Portland SD	59	49	77	61	67	43	73	57	45	59
Phoenix UHD	54	71	66	60	67	51	63	80	77	65
Boston PS	82	42	69	69	78	84	79	43	52	66
Clark County	48	75	67	50	43	49	56	55	81	58

RESULTING OUTCOMES — POST-SEGMENTATION DEPLOYMENT

**+23%**

Mean score improvement (Seg
A+C combined, 18-month
cohort)

↑ vs +9% baseline pre-
segmentation

**91%**

District contract renewal rate
after DAS rollout

↑ from 78% the prior year

**3.4×**

Faster intervention targeting via
ML segment classifier

↑ 12 hrs → 3.5 hrs per
counselor cycle

**\$4.2M**

Incremental ARR from upsell to
Seg B & D schools

↑ ROI on A/B program: 11.4× in
Y1

